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Optimizations in Hydraulic Fracturing Operations, the Impact of Using Dissolvable Frac Plugs in Saudi Arabia Fracturing Operations

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Abstract

Since 2017, dissolvable frac plugs have been deployed in Saudi Arabia for both unconventional and conventional development. Over 6,000 dissolvable plugs have been used, with 10% featuring acid-resistant coatings for acid fracturing. These plugs are categorized into two major types: Type 1, with a dual slip design capable of holding pressure from both sides, and another is Type 2, with a single slip design that enhances dissolution performance. This study also explores the applications of these plugs in acid and proppant fracturing operations.

This study analyzes the performance of dissolvable frac plugs in various well conditions across Saudi Arabia. The Type 1 plugs, with their dual slip design, were tested for their ability to maintain pressure integrity from both directions, while the Type 2 plugs were evaluated for their dissolution efficiency and drift rate. Data was collected from field deployments, focusing on parameters such as pressure holding capacity, dissolution time, and operational efficiency. Additionally, the impact of acid-resistant coatings on plug performance during acid fracturing was assessed. Proppant applications were also examined to determine the plugs' effectiveness in maintaining fracture conductivity and preventing proppant flowback.

The Type 1 plugs demonstrated excellent pressure rating up to 13,000 psi 30% increase, effectively holding pressure from both sides, which is crucial for maintaining well integrity during fracturing operations. The Type 2 plugs exhibited superior dissolution performance in 95 hours, ensuring no residual material obstructed the wellbore. Additionally, the acid-resistant coated plugs performed reliably in acid fracturing environments, preventing premature degradation. The collected data indicates that the use of dissolvable plugs significantly reduces the time and cost (milling time 3 days and cost reduction of 15%) associated with post-fracturing well interventions. By eliminating the need for coiled tubing milling or drifting, operators can transition from fracturing to production more swiftly, enhancing overall operational efficiency. This improvement not only optimizes the workflow but also contributes to cost savings and increased productivity, making dissolvable plugs a valuable innovation in hydraulic fracturing operations. The ability to move seamlessly from fracturing to production without additional interventions underscores the practical benefits of these advanced plugs, highlighting their role in modernizing and streamlining hydraulic fracturing processes.

The innovative use of dissolvable frac plugs in high-temperature, high-pressure (HTHP) wells present a significant advancement in hydraulic fracturing technology. The ability to flow back and produce directly without the need for coiled tubing interventions not only saves time and costs but also reduces operational risks. This approach accelerates the production timeline and offers a viable solution for wells with deformed casing, where traditional coiled tubing re-entry might be challenging.

Introduction

Hydraulic fracturing has revolutionized the extraction of hydrocarbons from both unconventional and conventional reservoirs, enabling significant advancements in resource development. Among the innovative technologies that have emerged in this field, dissolvable frac plugs have gained considerable attention for their ability to streamline operations and enhance efficiency. Since their introduction in 2017, over 6,000 dissolvable plugs have been deployed in Saudi Arabia, with approximately 10% featuring acid-resistant coatings designed specifically for acid fracturing applications. These plugs are classified into two primary types: Type 1 plugs, which utilize a dual slip design to maintain pressure integrity from both sides, and Type 2 plugs, which employ a single slip design to improve dissolution performance.

The application of dissolvable frac plugs addresses several challenges inherent in traditional hydraulic fracturing methods. Recent research by [Wang et al. \(2023\)](#) highlights the advantages of using new dissolvable isolation tools that eliminate plug-and-perf operation challenges, thereby reducing completion costs and enhancing operational efficiency. Furthermore, [Ningjing and Qijun \(2017\)](#) discuss the significance of dissolvable tools in multistage stimulation operations, underscoring their potential to optimize workflows and minimize the need for additional interventions.

Recent studies have also explored the use of acid-based crosslinked gels to improve fluid dynamics during acid fracturing processes. Almulhem, [Malik, and Ghazwi \(2022\)](#) demonstrated that these gels enhance fluid flow and distribution in carbonate formations, which is critical for maximizing the effectiveness of fracturing operations. Additionally, the work of [Xu et al. \(2015\)](#) emphasizes the importance of spontaneously imbibed fluids in increasing the contact area between hydraulic fracturing fluids and the formation matrix, particularly in liquids-rich shale plays. Such advancements are crucial for ensuring efficient resource recovery and minimizing operational risks.

Moreover, the optimization of fluid laboratory testing for conventional proppant fracturing applications in high-temperature reservoirs, as discussed by [Yudin et al. \(2024\)](#), further illustrates the need for adaptive methodologies to enhance operational performance under challenging conditions. The integration of dissolvable frac plugs into hydraulic fracturing practices not only addresses the limitations of traditional techniques but also contributes to a more sustainable and cost-effective approach to resource extraction.

This study aims to evaluate the performance of dissolvable frac plugs in various well conditions across Saudi Arabia, focusing on their ability to maintain pressure integrity, improve dissolution efficiency, and enhance overall operational effectiveness during both acid and proppant fracturing operations. By analyzing key performance indicators, this research will provide valuable insights into the impact of dissolvable plugs on hydraulic fracturing methodologies, further establishing their role as a transformative innovation in the field.

Reservoir and Execution Challenges Associated with Plug Implementation

As the global energy landscape evolves, unconventional oil and gas reservoirs—particularly those located in high-temperature, high-pressure (HTHP) environments—are playing an increasingly vital role ([Xu, L 2015](#)). These reservoirs often exist in ultra-deep formations, characterized by bottomhole temperatures exceeding 250°F (121°C) and differential pressures surpassing 10,000 psi. The advancement of multistage hydraulic fracturing technologies has enabled access to these resources, with plug-and-perf systems being

the predominant completion method. However, the effectiveness of this method hinges critically on the performance of frac plugs under such extreme downhole conditions.

Dissolvable frac plugs are favored for their post-operation dissolving, which reduces intervention costs and minimizes risks associated with debris left in the wellbore. Traditional plugs, which often incorporate elastomeric (rubber) elements for sealing, are commonly used in HTHP environments. The failures are often magnified by the inability of these materials to withstand over 10,000 differential pressure in deep unconventional plays.

One of the primary technical challenges is achieving a predictable dissolution profile in downhole conditions. Magnesium-aluminum alloys, widely used in dissolvable plug designs, dissolve rapidly at high temperatures. This behavior, while useful in moderate environments, becomes a liability in HTHP settings, where premature dissolution before the fracturing operation is complete can jeopardize zonal isolation. Additionally, the dissolution behavior is influenced by chloride ion concentration, pH of the fracturing fluid, fluid velocity, and formation pressure—all of which vary across wells.

Mechanical challenges are also prevalent. The plug must be structurally sound to hold differential pressures above 10,000 psi and resist deformation during setting and fracturing. The plug's components—such as slips, seal rings, cones, and support subs—must function cohesively to achieve anchoring and isolation. Misalignment during setting or deformation under stress can lead to failure modes such as slip collapse, seal extrusion, or plug ejection.

Moreover, operational practices, including setting depths, timing between plug setting and fracturing, and pressure cycling, introduce variability. Engineers must consider how long the plug needs to last before initiating dissolution. A common target is a 24-hour stability window, during which the plug must maintain full functionality.

The lack of field-specific research data and real-time feedback from downhole sensors makes plug design and material selection highly iterative and conservative. These uncertainties necessitate robust testing, empirical modeling, and material innovation to ensure safety and cost-efficiency.

Solutions and Statistical Analyses

Material Selection and Characterization

Material selection is critical to plug reliability and dissolution performance. Magnesium-aluminum alloys were evaluated based on their dissolution rates (Ningjing, J 2017), tensile strength, elongation, and corrosion resistance. Under controlled laboratory conditions (350°F, 2% KCl solution), seven materials were tested. Key findings are presented in the chart (Fig. 1) below:

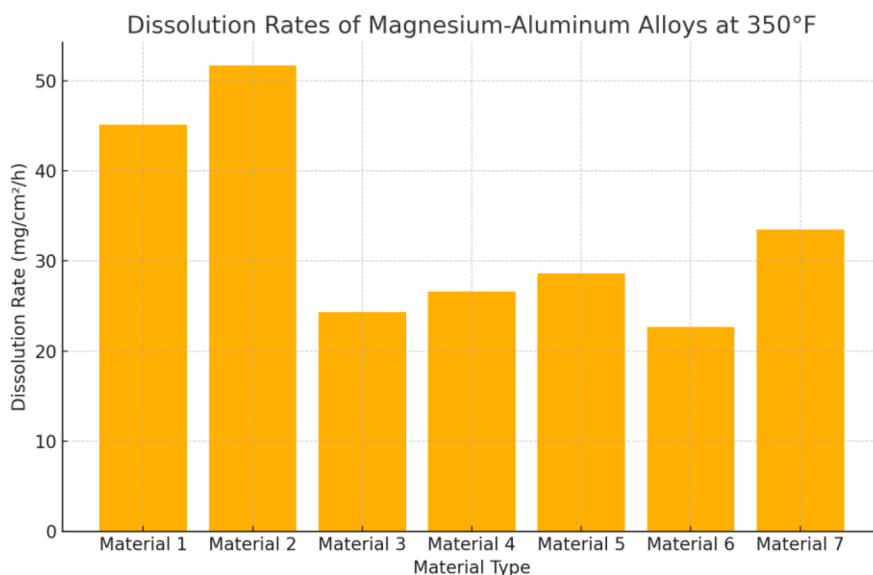
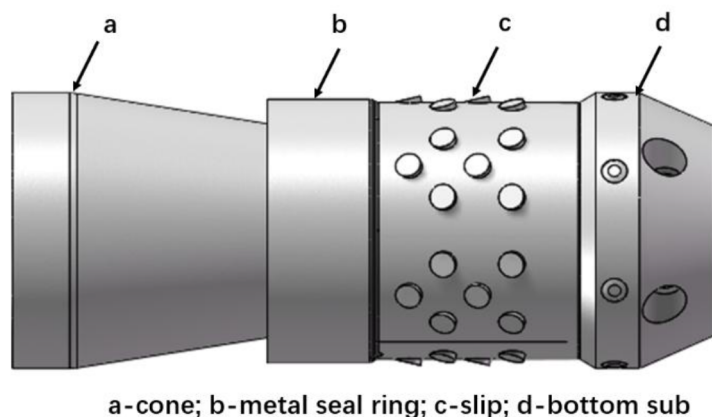


Figure 1—Rate of Dissolution Chart

Materials 3, 6, and 7 demonstrated the best balance between strength and dissolution control. Material 6, in particular, provided superior performance in high-chloride environments, maintaining mechanical integrity beyond 24 hours and displaying uniform dissolution patterns.



a-cone; b-metal seal ring; c-slip; d-bottom sub

Figure 2—Plug Structure Demonstration

Structural Optimization

Advanced CAD and FEA modeling were employed to refine the plug's mechanical architecture. Key innovations included:

- Reinforced slip designs to prevent buckling under axial load.
- High-ductility seal rings that deform predictably during setting, ensuring casing contact without fracturing.
- Cone angle optimization for improved load transfer.
- Reduction of internal plug volume to increase flow-through diameter while maintaining structural robustness.

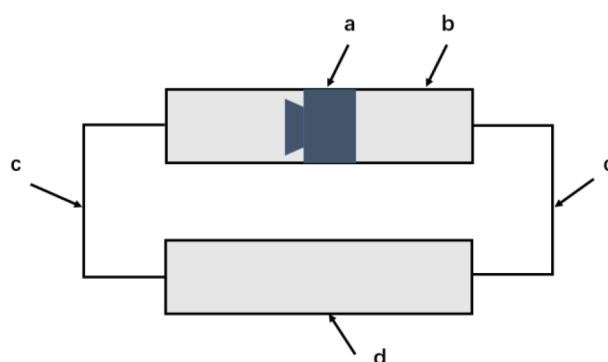
Stress simulations showed improved resistance to casing eccentricity and better accommodation of thermal expansion. These enhancements directly translated into field reliability.

Lab Test Validation Pressure Holding and Thermal Stability Testing

A simulated HTHP test regime was implemented:

1. Pre-soaking plugs at 350°F in brine for 24 hours.
2. Pressure testing at 15,000 psi for 5 hours using 2% KCl solution.
3. Dissolution monitoring over a 96-hour period.

Under the condition of 350°F, the HTHP frac plug withstands a pressure of 15,000psi and maintains this pressure for 5 hours after 24 hours pre-soaking, showing no sudden pressure decline and meeting the requirements for high-temperature and high-pressure resistance. The test system and the pressure-holding curve of the frac plug are shown in Fig. 3 and Fig 4.



a-plug after set; b-casing; c-high pressure test line;
d-pressure control equipment

Figure 3—Diagram of Test System for HTHP Metal Seal Dissolvable Frac Plug

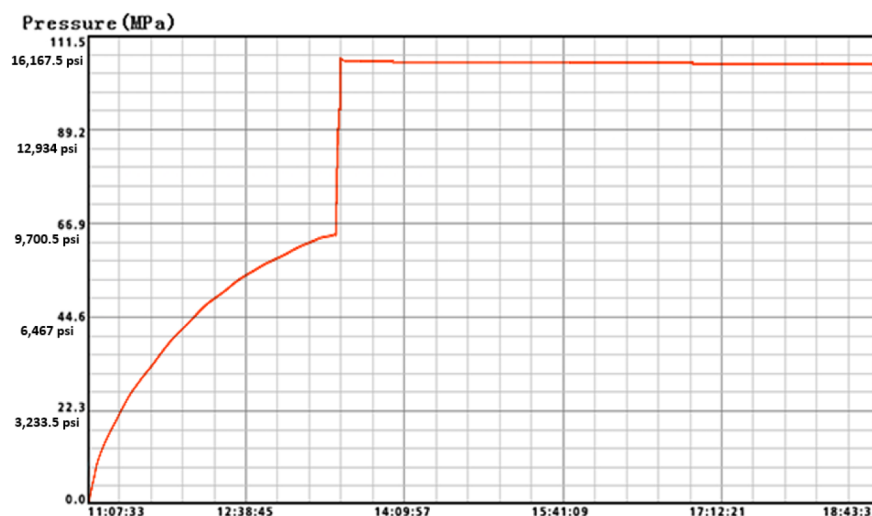


Figure 4—Pressure Holding Curve of HTHP Metal Seal Dissolvable Frac Plug

Pressure Cycle Lab Validation Simulating Frac Pressure Cycles

Moreover, the metal seal design increases the plug pressure holding ability, more importantly it has better ability to handle more than 25 times of HTHP cycles (as shown in Fig. 5), and it helps to handle the multiple pressure cycles of different fracturing phases and also adds up the safety margin when fracturing the tight formation.

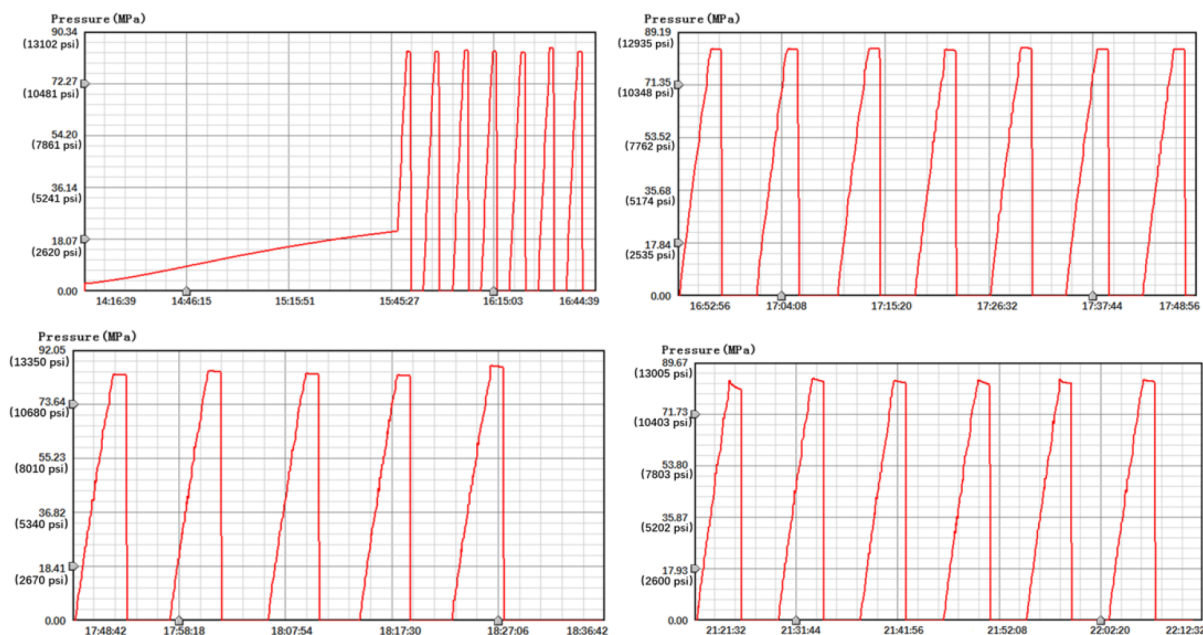


Figure 5—HTHP Pressure Cycles

Environmental and Operational Benefits

Dissolvable plugs eliminate the need for post-fracturing milling, reducing coiled tubing deployment time by up to 40%. They also minimize wellbore debris, lowering the risk of tool sticking and improving well integrity. From an ESG standpoint, the dissolvable system reduces fluid consumption and waste generation, aligning with industry sustainability targets.

Case Studies

Overview

The following case studies will present two wells in which isolation between stages solely utilized dissolvable frac plugs. Well A targeted a sandstone formation in which proppant fracturing treatments were performed. In contrast, Well B was stimulated with a series of acid fracturing treatments in a carbonate formation. In both cases, the plugs were exposed to high-pressure and high-temperature conditions without experiencing any performance failures. The wells were able to be opened for flowback after concluding the fracturing operations without the requirement of coiled tubing milling highlighting a significant improvement in operational efficiency over conventional frac plugs.

Well A

Formation and Well Completions. Well A was drilled as an S-Shaped well targeting a gas producing sandstone formation. The well was completed with a 4.5" production tubing and cemented liner. Reservoir pressure was estimated to be 5,000 psi while formation temperature representing the maximum temperature to which the frac plugs might be exposed was 310°F. Each stage was perforated with a single 5 ft cluster consisting of 4 shots per foot. Three proppant fracturing treatments were designed to cover the target interval across of 13,150 – 13,450 ft TVD (14,675 – 14,975 ft MD).

Treatment Design. Reservoir conditions were taken into consideration for treatment fluid design and proppant requirements. Crosslinked gelled fluid utilizing CMHPG gelling agent and zirconate crosslinker was selected due to the high reservoir temperature. The fluid formulation was customized based on the stage pump time in order to achieve the required gel stability under downhole conditions. A combination of

30/50 and 20/40 mesh size proppant was used consistent with offset wells in the area. Due to the formation characteristics, HSP (high strength proppant) was required followed by RCP (resin coated proppant) at the end of the treatment in order to mitigate potential sand production. The treatments were designed for 175,000 – 220,000 lbs total proppant up to a maximum proppant concentration of 6.5 PPA. Below is an example treatment schedule.

Table 1—Well A Average Pumping Schedule

STG #	PROPPANT TYPE	FLUID TYPE	PUMP RATE (BPM)	CLEAN VOLUME (GAL)		PROPPANT CONCENTRATION (PPA)	PROPPANT MASS (LB)	
			DIRTY	STAGE	CUM.	STAGE	STAGE	CUM.
1	Pad	Crosslinked 45#	40	30,000	30,000			
2	Sinterblast	Crosslinked 45#	40	6,000	36,000	0.5	3,000	3,000
3	30/50 HSP	Crosslinked 45#	40	4,000	40,000	1	4,000	7,000
4	30/50 HSP	Crosslinked 45#	40	5,500	45,500	2	11,000	18,000
5	30/50 HSP	Crosslinked 40#	40	7,000	52,500	3	21,000	39,000
6	30/50 HSP	Crosslinked 40#	40	7,000	59,500	4	28,000	67,000
7	30/50 HSP	Crosslinked 40#	40	9,000	68,500	4.5	40,500	107,500
8	20/40 RCP	Crosslinked 40#	40	8,500	77,000	4.5	38,250	145,750
9	20/40 RCP	Crosslinked 40#	40	4,000	81,000	5	20,000	165,750
10	20/40 RCP	Crosslinked 40#	40	1,500	82,500	6	9,000	174,750
11	Flush	Linear Gel 20#	40	9,000	91,500			

Treatment Execution. The three stage proppant fracturing operations were executed over the course of five days. Diagnostic pumping tests including injection tests, step rate tests, and data fracs were also performed in order to optimize the treatment design for each stage. Logistics for frac fluid chemicals, water, and proppant was very crucial to the operation. Any delays that happen to occur once the dissolvable plug has entered the well increase the risk of failure while pumping. Because of this, product logistics was planned to ensure all requirements were available at the wellsite before beginning the plug and perforation operations for the stage.

All stages were executed successfully as per design. Average proppant mass pumped per stage was 200,000 lbs while average fluid volume pumped was 2,700 bbl. Average pumping rate during the treatments was 40 bpm. The pumping time for the proppant fracturing treatment of each stage was approximately one hour. As mentioned, diagnostic pumping tests were performed prior to the main treatment each of which lasting approximately 10-20 min. Below is a treatment chart from one of the stages showing treating pressure, rate, and proppant concentration.

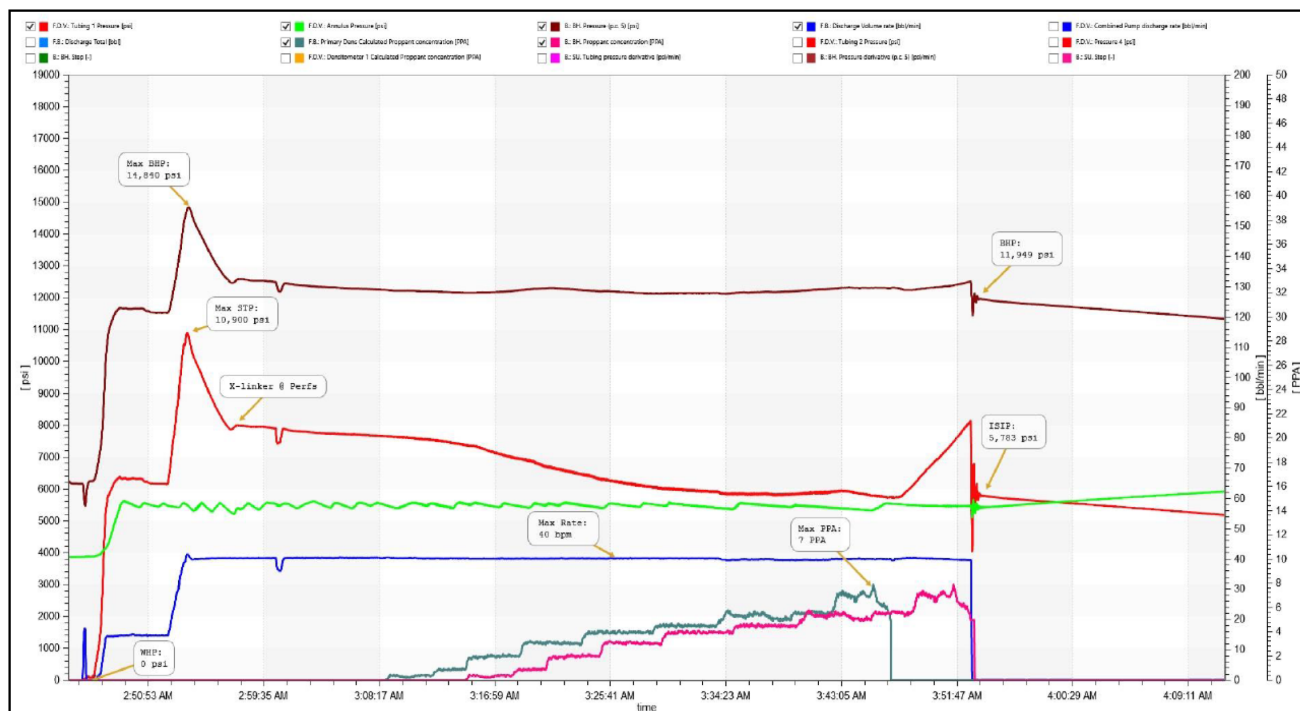


Figure 6—Well A Proppant Fracturing Treatment Plot

As previously highlighted, stage timing is critical to ensuring the integrity of the plug during the pumping operations. A plug failure not only results in a failed fracturing treatment, but additional intervention and a refrac of the stage might be required leading to lost time and significantly reduced operational efficiency. Therefore, it is important to minimize the time between first exposure of the plug to the wellbore fluid and completion of the main proppant fracturing treatment. For the plug isolating stage 1 (stage 2 plug and perforation), the time between the start of wireline running in the well to the completion of the main fracturing treatment was 28 hours. In addition to being exposed to bottom hole conditions, during that time the plug also experienced three pressure cycles (Injection Test, Step Rate Test, and the main fracturing treatment) with average pressure of 12,000 psi and maximum pressure of 15,000 psi.

Similarly, stage 3 fracturing operations concluded 29 hours after the plug had entered the wellbore. Again, an Injection Test and Step Rate Test were performed prior to the main treatment. Maximum pressure the plug was exposed to during the entire stage operations was 15,500 psi with average pressure of 12,000 psi. The results from both stages demonstrate the plug's ability to successfully handle the HPHT condition requirements even after 24 hours of exposure.

Well B

Formation and Well Completions. Well B is a horizontal gas well targeting a carbonate formation. Similarly, a 4.5" production tubing with cemented liner was used to the well completion. The reservoir temperature and pressure are 280°F and 4,500 psi. The total MD of the well was 18,000 ft with average TVD across the perforation intervals of 12,000 ft. A total of eight acid fracturing stages were planned. Between five to seven perforation clusters were used per stage each of which were 2 ft. Total stage length ranged between 230 – 650 ft.

Treatment Design. Due to the depleted nature of the formation, the stimulation treatments were designed with CO₂ foamed fluids targeting an average foam quality of 60%. CO₂ aids in fracturing fluid flowback especially in under pressured reservoirs. It is pumped as a liquid but converts to gas with increasing temperature and the corresponding reduction in pressure during flowback operations. Care must be taken in

proper fluid selection as CO₂ is not compatible with all types of fluids. For the treatment design, 26% HCl along with linear gelled and crosslinked acids (each consisting of 15% HCl) were used. These gelled acids aid in establishing frac geometry, contribute to enhanced fluid diversion across the target interval, and have a lower reaction rate allowing for deeper penetration into the formation. Sufficient corrosion inhibition and iron control was included in the acid formulations based on the well conditions.

Stage treatments were designed with several cycles of the above-mentioned three acid fluid systems. Each cycle started with crosslinked acid for breaking down the formation and establishing the frac geometry followed by 26% HCl and linear gelled acid. Additionally, solid diverter was included in alternating cycles in order to further improve fluid distribution across the target interval. The number of cycles and acid volume per stage varied depending on the stage length with six to seven cycles and 800 to 1100 bbl total acid being pumped along with an average of 165 tonnes of CO₂ per stage. The target downhole rate was 30-35 bpm inclusive of both treatment fluid and CO₂ combined. A 75% wellbore overflush with treated water was performed at the end of each treatment. It was crucial to ensure that the foamed fluids were fully displaced into the formation to avoid any complications with subsequent wireline plug and perforate operations, except for the final stage where CO₂ would further aid in flowing back the well. Below is an example of the typical treatment schedule that was pumped.

Table 2—Well B Average Pumping Schedule

Step #	Stage Name	Surface Calculations						Bottomhole Calculations				
		Total Pump Rate	Liquid Rate	CO ₂ Rate	Fluid Name	Clean Stage Volume (non CO ₂)	CO ₂ Vol	Slurry Stage Volume	Foam Quality	Liquid rate	CO ₂ Rate	Total Rate
		[bpm]	[bpm]	[bpm]		[gal]	[tonne]	[gal]	[%]	[bpm]	[bpm]	[bpm]
1	Cooldown stage	10	10	0	Treated Water	3,150	0.0	3,150	0	10	0	10
2	Acid-0	12	12	0	HCl 26%	2,000	0.0	2,000	0	12	0	12
3	Acid Displacement	12	12	0	Treated Water	11,000	0.0	11,000	0	12	0	12
4	Pad-1	15	15	0	Crosslinked Gelled Acid	3,000	0.0	3,000	0	15	0	15
5	Acid-1	15	15	0	HCl 26%	6,000	0.0	6,000	0	15	0	15
6	Diversion-1	15	15	0	Linear Gelled Acid	2,000	0.0	2,000	0	15	0	15
7	Pad-2	20	8	12	Crosslinked Gelled Acid	1,400	7.8	3,500	60	8	12	20
8	Acid-2	20	8	12	HCl 26%	2,400	13.4	6,000	60	8	12	20
9	Spacer	15	6	9	Linear Gel	200	1.1	500	60	6	9	15
10	Diversion-2	15	6	9	RAD	200	1.1	500	60	6	9	15
11	SPacer	15	6	9	Linear Gel	200	1.1	500	60	6	9	15
12	Pad-3	20	8	12	Crosslinked Gelled Acid	1,400	7.8	3,500	60	8	12	20
13	Acid-3	26	11	16	HCl 26%	2,400	13.4	6,000	60	11	16	27
14	Diversion-3	26	11	16	Linear Gelled Acid	800	4.5	2,000	60	11	16	27
15	Pad-4	26	11	16	Crosslinked Gelled Acid	1,400	7.8	3,500	60	11	16	27
16	Acid-4	26	11	16	HCl 26%	2,400	13.4	6,000	60	11	16	27
17	Spacer	15	6	9	Linear Gel	200	1.1	500	60	6	9	15
18	Diversion-4	15	6	9	RAD	200	1.1	500	60	6	9	15
19	SPacer	15	6	9	Linear Gel	200	1.1	500	60	6	9	15
20	Pad-5	26	11	16	Crosslinked Gelled Acid	1,400	8.6	3,500	60	11	16	27
21	Acid-5	29	12	17	HCl 26%	2,400	14.8	6,000	60	12	18	30
22	Diversion-5	29	12	17	Linear Gelled Acid	800	4.9	2,000	60	12	18	30
23	Pad-6	29	12	17	Crosslinked Gelled Acid	1,400	7.8	3,500	60	12	18	30
24	Acid-6	29	12	17	HCl 26%	2,400	13.4	6,000	60	12	18	30
25	Overflush	30	30	0	Treated Water	7,875	0.0	7,875	0	30	0	30
26	Flush	30	30	0	Treated Water	10,500	0.0	10,500	0	30	0	30

Treatment Execution. The fracturing treatments for the eight stages were completed within twelve days. Again, product logistics was crucial for ensuring the success of the operation and reducing the time each plug was exposed to wellbore conditions before execution of the acid fracturing stimulation. In addition to the delivery and wellsite inventory management of raw HCl, water, chemicals, and CO₂, all fluid systems were mixed onsite which also needed to be considered in the operations plan.

Another critical aspect to the efficiency of the operation is the successful execution of the wireline plug and perforation runs for each stage, especially in horizontal wells. Because of this, wireline pump down

technique was chosen over traditional tractor method. This technique involves pumping the bottom hole assembly to the target depth at low rate once it enters the lateral section of the well. As a result, the average time per stage to complete the plug and perforation runs was 7 hours. This reduced the dissolvable plug exposure time in the wellbore to 18 hours on average up to completion of the main acid fracturing treatment.

Prior to each acid fracturing stimulation, an injection test was performed in order to confirm formation injectivity. This again demonstrates the ability of dissolvable plugs to maintain integrity over multiple pressure cycles. If a minimum injection rate cannot be achieved within the designed pressure limitations, acid spearhead treatments with coiled tubing are commonly performed to aid in breaking down the formation. In the case of Well B, coiled tubing intervention was not required in any of the stages.

The average stage pumping time was 2 hours. During this time, the plugs were exposed to an average bottom hole pressure of 13,000 psi with maximum pressure of over 17,000 psi. The average downhole pumping rate was 33 bpm while the maximum was 35 bpm. Fig. 7 below is the treatment chart from one of the stages showing the typical pressure and rate parameters during the operation.

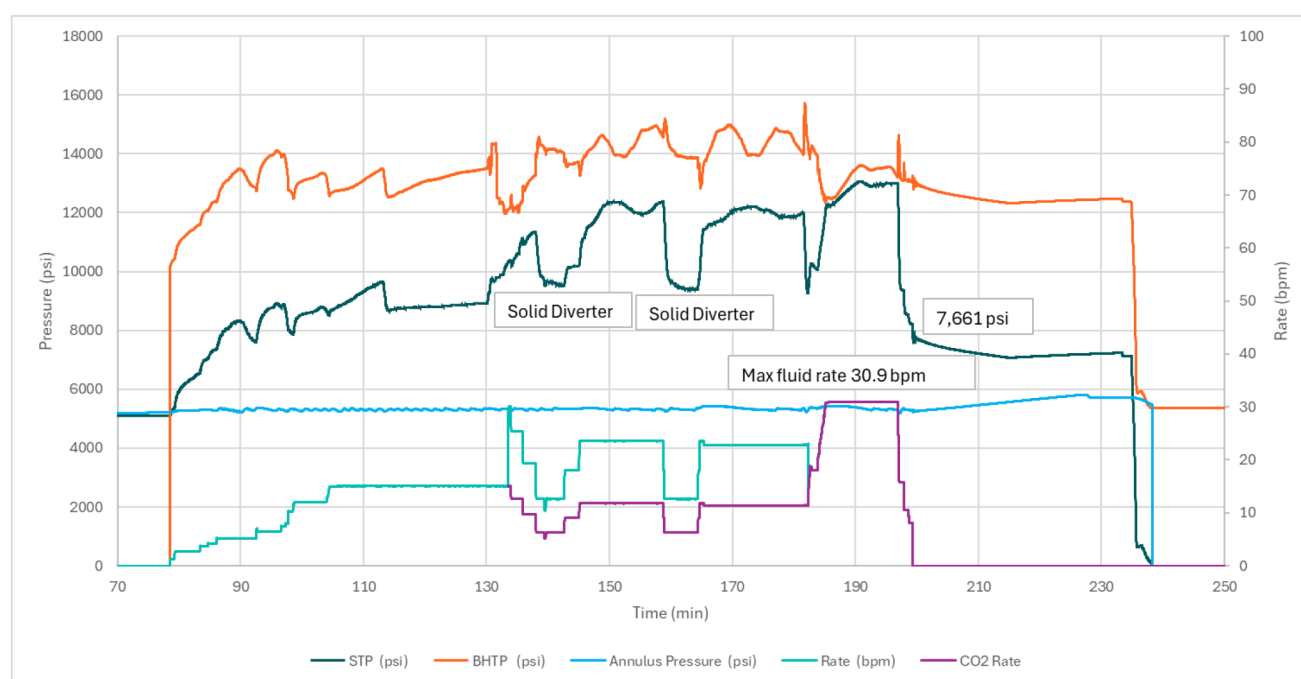


Figure 7—Well B Acid Fracturing Treatment Plot

Conclusion

The introduction of dissolvable frac plugs in hydraulic fracturing operations marks a significant advancement in the efficiency and effectiveness of well completions in Saudi Arabia. This study highlights the profound impact of these plugs on operational workflows, particularly in high-temperature and high-pressure environments. The performance evaluation of Type 1 and Type 2 dissolvable plugs demonstrates their ability to maintain pressure integrity, optimize dissolution rates, and enhance overall operational efficiency. The findings indicate that the use of dissolvable frac plugs not only minimizes the time and costs associated with post-fracturing interventions but also significantly improves production timelines by eliminating the need for coiled tubing milling. This innovation not only streamlines the transition from fracturing to production but also reduces operational risks associated with residual materials in the wellbore. Furthermore, the successful application of dissolvable plugs in both proppant and acid fracturing treatments across various well conditions underscores their versatility and reliability. The ability to operate effectively under extreme downhole conditions, coupled with reduced environmental impact through minimized fluid

consumption and waste generation, aligns with industry sustainability goals. In conclusion, the deployment of dissolvable frac plugs represents a critical step forward in hydraulic fracturing technology, offering substantial benefits in terms of efficiency, cost savings, and enhanced well integrity. Continued research and development in this area will further optimize their design and application, paving the way for more innovative solutions in the evolving energy landscape.

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